The Research Lifecycle in Three Supply Chain Shortage Studies

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The research lifecycle runs from identifying a problem through design, data collection, analysis, interpretation, and dissemination, with ethical obligations carried through every stage. This paper traces that lifecycle in three studies of supply chain shortage and access, a topic central to my proposed study, *Mitigation of Supply Chain Uncertainties Caused by Material Shortage During Ramp-Up*, which examines semiconductor manufacturing equipment providers and their suppliers. The qualitative approach is illustrated by Mlyuka et al. (2022), the quantitative approach by the analytical model of Kouvelis et al. (2023), and the mixed methods approach by Asamoah et al. (2021). Three further studies, Namisango et al. (2017), Singh et al. (2026), and Bimha et al. (2019), provide supporting comparisons. Table 1 and Figure 1 summarize the three lifecycles.

Qualitative Approach: Mlyuka et al. (2022)

Research Problem, Purpose, and Questions

Mlyuka et al. (2022) addressed the limited availability of hydroxyurea, the main disease-modifying medicine for sickle cell disease, in Tanzania, even though Sub-Saharan Africa carries about 75% of the global disease burden. No Tanzanian manufacturer produced the drug, and only one imported brand was registered. The purpose was to explore the barriers to and facilitators of hydroxyurea availability from the perspective of the upstream supply chain. The guiding question asked what prevents and what enables manufacturers, importers, and regulators to make the medicine available. Consistent with exploratory qualitative inquiry, the study stated no hypotheses.

Methodology and Design

A qualitative methodology fit the problem because the barriers lay in perceptions, business judgments, and regulatory understanding rather than in measurable variables. The design was a qualitative case study with purposive sampling of 10 participants who each had more than three years of experience in pharmaceutical manufacturing, importation, or regulation (Mlyuka et al., 2022).

Research Instruments and Analysis

The instrument was a semistructured interview guide used in in-depth interviews, with recruitment continuing until data saturation. Interviews were audio-recorded and transcribed verbatim, and the transcripts were analyzed thematically, producing two themes: barriers and opportunities (Mlyuka et al., 2022).

Key Findings

Inadequate understanding of sickle cell disease, hydroxyurea, and the national orphan drug regulation heightened concern about a small market and weak investment viability, which discouraged local manufacturing and importation. Participants nonetheless viewed the regulator as supportive and saw existing policies as a starting point for improving availability (Mlyuka et al., 2022; see Figure 2). Namisango et al. (2017) reached a similar conclusion for opioids in Mozambique, Swaziland, and Zimbabwe, where document review and group interviews revealed restrictive laws, inaccurate estimation of consumption, and a shortage of prescribers (see Figure 3).

Ethical and Legal Implications

Interviewing a small number of identifiable industry and regulatory professionals requires ethical clearance, informed consent, and careful de-identification, because commercial strategies

and regulatory opinions are sensitive. Legally, the findings turn on how orphan drug and medicines registration rules are understood and applied; Namisango et al. (2017) likewise showed that national narcotics laws shaped the opioid supply chain, so researchers must report legal barriers accurately without encouraging noncompliance.

Quantitative Approach: Kouvelis et al. (2023)

Research Problem, Purpose, Questions, and Hypotheses

Kouvelis et al. (2023) addressed material shortages in project supply chains, where an engineer-to-order or make-to-order firm depends on a contract supplier for a key material while facing uncertain activity times and uncertain material consumption. The purpose was to determine how the firm should combine inventory, time buffers, and supplier flexibility. The research questions asked how the firm should jointly set its promised due date and advance material order, and how contract terms allocate shortage risk between firm and supplier. In place of statistical hypotheses, the analytical model produced formal propositions about optimal decisions.

Methodology and Design

The methodology was quantitative and deductive. The design was an analytical stochastic model of a project firm and its supplier operating under a flexible wholesale price contract with a discounted advance order price and a higher expedite price (Kouvelis et al., 2023). Singh et al. (2026) illustrate the same quantitative logic in a humanitarian setting by building a shortage mechanism and realistic constraints directly into an optimization model.

Research Instruments and Analysis

In modeling research, the instrument is the mathematical model: decision variables (due date and order quantity), random variables (activity time and material consumption, which may

be positively or negatively correlated), and cost parameters such as late-delivery penalties. Kouvelis et al. (2023) analyzed the model by deriving optimal policies and examining how they change as uncertainty, penalties, and contract prices change (see Figure 4).

Key Findings

The due date and the material order generally act as substitutes, so a longer time buffer can replace inventory and the reverse. Greater uncertainty always increases material orders but may lengthen or shorten the due date. The discounted advance price shifts inventory risk to the firm, while the expedite price leads the supplier to hold safety stock (Kouvelis et al., 2023).

Ethical and Legal Implications

Modeling studies involve no human participants, but they carry ethical duties of transparency: assumptions, parameter values, and limits must be stated so that managers do not over-apply results. Legally, the findings depend on contract terms such as penalty clauses and expedite pricing, which must be enforceable and fairly negotiated, particularly when the supplier has more bargaining power.

Mixed Methods Approach: Asamoah et al. (2021)

Research Problem, Purpose, and Questions

Asamoah et al. (2021) addressed gaps in how health care providers maintain the vaccine cold chain, where temperature failures destroy supply and endanger patients. The purpose was to assess providers' knowledge, attitudes, practices, and challenges in cold chain management. The questions asked what providers know and believe, how they practice, whether knowledge and attitudes are related, and what obstacles they face.

Methodology and Design

The methodology was mixed methods with a concurrent design at 12 health facilities in the Sekyere Central District of Ghana (Asamoah et al., 2021). Quantitative and qualitative data were collected in the same period and integrated at interpretation. Bimha et al. (2019) used a comparable concurrent parallel design to study supply chain management practices and competitiveness in the Zimbabwean petroleum industry, choosing it for its robustness in complex supply chain settings.

Research Instruments and Analysis

The instruments were a survey questionnaire completed by 86 providers, an observation checklist applied at 12 cold chain sites, and interview guides for 11 key informants (see Figure 5). Quantitative data were analyzed with descriptive statistics and correlation analysis, and interview data were coded inductively into themes (Asamoah et al., 2021).

Key Findings

Most participants showed good knowledge (68.6%) and good attitudes (67.4%), but the relationship between knowledge and attitudes was weak. Interviews identified inadequate personnel, erratic power supply, logistical constraints, and transportation difficulties as major challenges, and the authors recommended continuing education, adequate resources, and stronger monitoring (Asamoah et al., 2021).

Ethical and Legal Implications

Studying health workers requires ethical approval, informed consent, and assurance that findings about poor practice will not be used against individuals. Observation raises added duties to avoid disrupting care and to report unsafe storage. Legally, vaccine handling is governed by

national immunization regulations, so findings imply compliance gaps that facilities must address.

Comparative Synthesis and Implications for the Proposed Study

The research lifecycle changes with methodology. Mlyuka et al. (2022) built understanding inductively from the experience of supply chain actors, Kouvelis et al. (2023) deduced optimal decisions from a formal model, and Asamoah et al. (2021) combined measurement with lived experience. The findings converge: shortages arise from weak estimation and visibility, uncertain market signals, rigid or poorly understood rules, limited capacity, and fragile logistics, and they are reduced by buffers, flexible supplier terms, and monitoring.

For my proposed qualitative case study, Mlyuka et al. (2022) offer a direct model: purposive sampling of experienced upstream actors, in-depth interviews to saturation, and thematic analysis, which Namisango et al. (2017) show can be strengthened by document review; Kouvelis et al. (2023) and Singh et al. (2026) provide constructs and analytics for interpreting inventory, time buffers, and supplier flexibility during ramp-up; and Asamoah et al. (2021) show how observation can verify what participants report. Because semiconductor firms operate under nondisclosure agreements and export controls, IRB approval, informed consent, de-identification of firms, and careful screening of proprietary documents will be essential. The Appendix translates these findings into a shortage risk-scoring algorithm.

Table 1

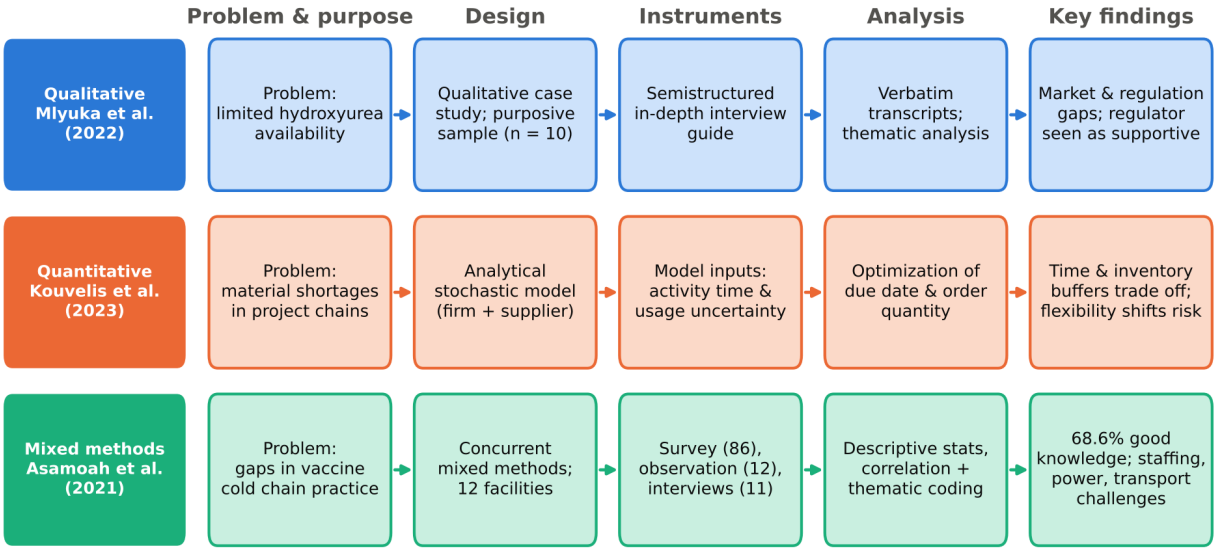
Comparison of the Research Lifecycle Across the Three Studies

Element	Mlyuka et al. (2022)	Kouvelis et al. (2023)	Asamoah et al. (2021)
Method	Qualitative (case study)	Quantitative (analytical modeling)	Mixed methods (concurrent)
Problem	Limited hydroxyurea availability in Tanzania	Material shortages in engineer-to-order project supply chains	Gaps in vaccine cold chain practice
Questions / hypotheses	What hinders and enables upstream availability; no hypotheses	Optimal due date and material order; formal propositions	Knowledge, attitudes, practices, challenges; relationship tests
Design	Case study; purposive sample of 10 experienced actors	Stochastic model of firm and contract supplier	Concurrent design at 12 facilities
Instruments	Semistructured in-depth interview guide	Mathematical model and parameters	Survey (86), observation checklist (12 sites), interviews (11)
Analysis	Verbatim transcription; thematic analysis (2 themes)	Derivation of optimal policies; sensitivity	Descriptive statistics, correlation, thematic coding
Key findings	Awareness and regulation gaps; market viability concerns; supportive regulator	Time and inventory buffers substitute; contract terms shift risk	68.6% good knowledge; staffing, power, transport gaps
Ethics / legal	Consent, de-identification, orphan drug regulation	Transparency of assumptions; contract law	Consent, observation duties, immunization rules

Note. Compiled by the author from Mlyuka et al. (2022), Kouvelis et al. (2023), and Asamoah et al. (2021). Column colors match Figure 1: blue = qualitative, orange = quantitative, green = mixed methods.

Figure 1

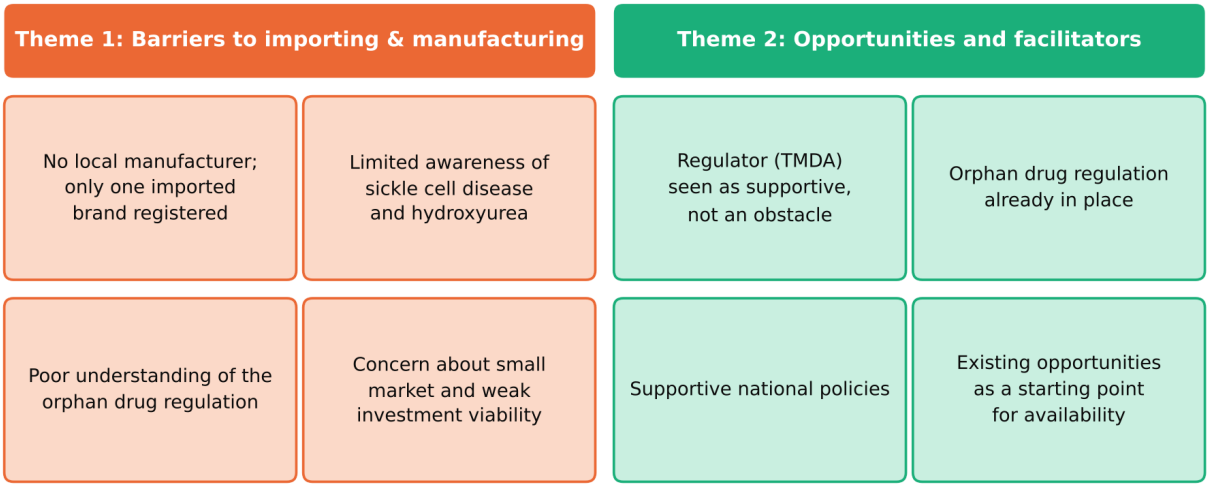
Research Lifecycle Stages in the Qualitative, Quantitative, and Mixed Methods Studies



Note. Each row traces one study from problem to findings. Compiled from Mlyuka et al. (2022), Kouvelis et al. (2023), and Asamoah et al. (2021).

Figure 2

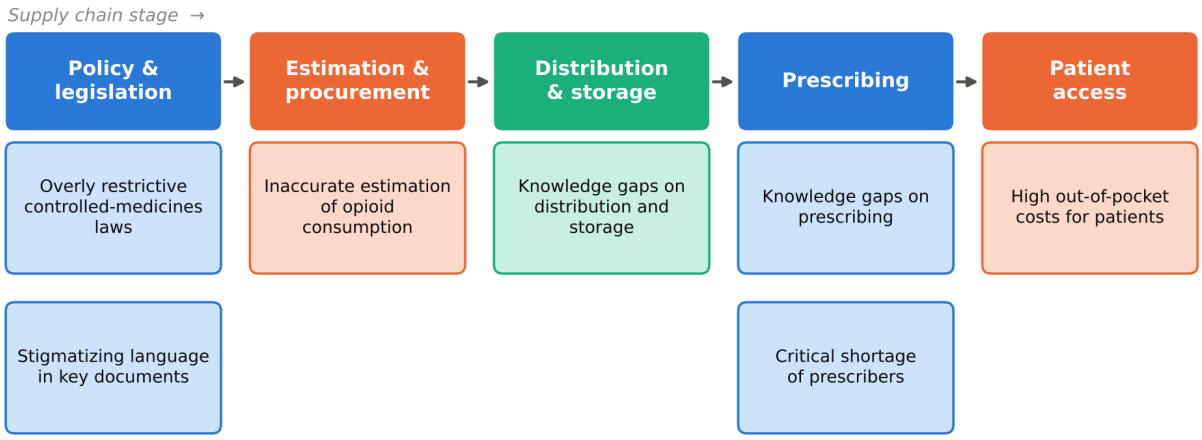
Themes on Barriers to and Facilitators of Hydroxyurea Availability



Note. Themes reported by Mlyuka et al. (2022) from in-depth interviews with pharmaceutical manufacturers, importers, and regulators. TMDA = Tanzania Medicines and Medical Devices Authority.

Figure 3

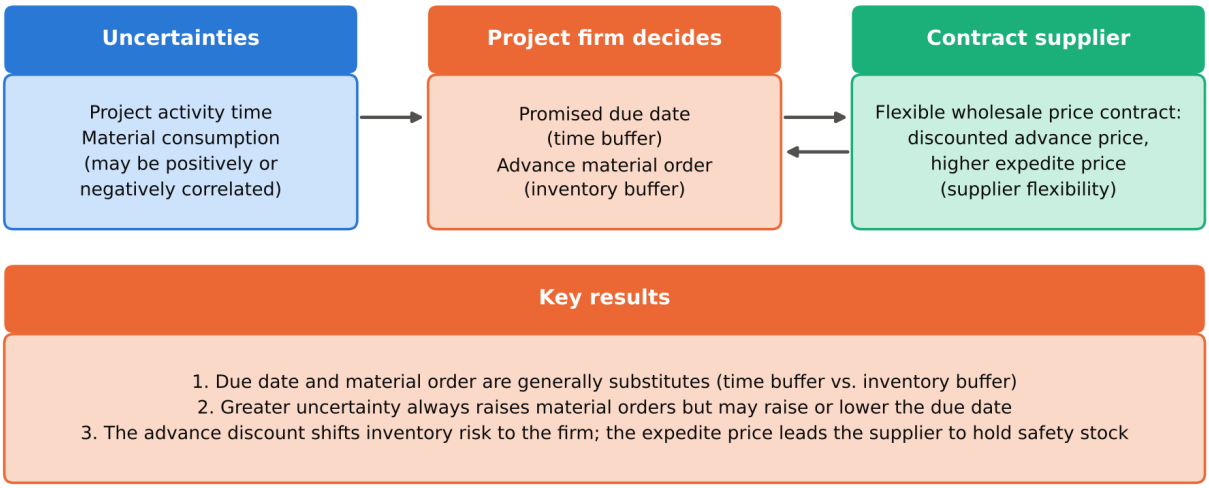
Barriers to Opioid Availability Mapped to Supply Chain Stages



Note. Barriers reported by Namisango et al. (2017), arranged by the author along the supply chain from policy to the patient.

Figure 4

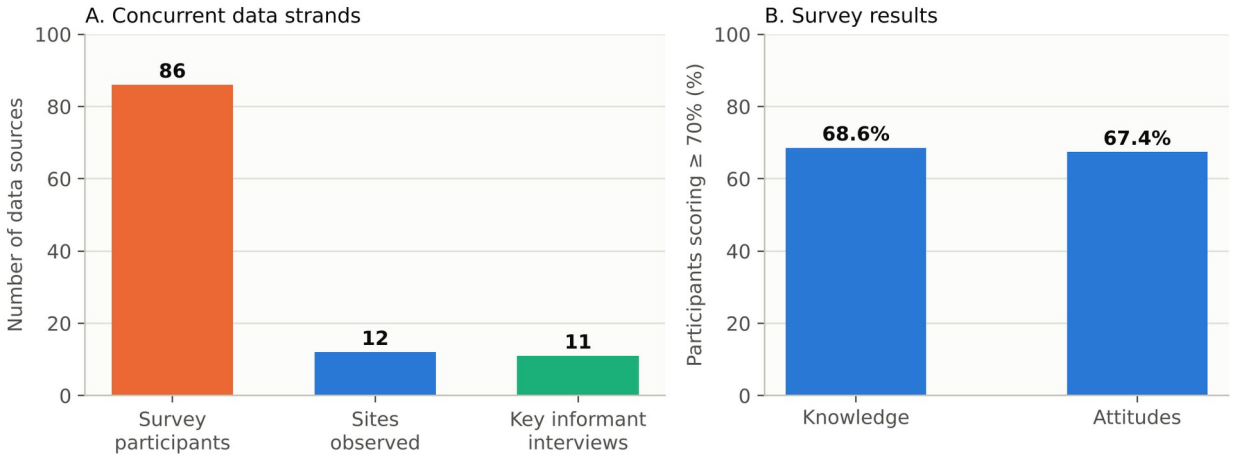
Structure and Key Results of the Project Supply Chain Shortage Model



Note. Summarized from Kouvelis et al. (2023). The three buffers in the model, inventory, time, and supplier flexibility, correspond to mitigation strategies in the proposed study.

Figure 5

Concurrent Data Strands and Survey Results in a Mixed Methods Cold Chain Study



Note. Panel A shows the three data sources collected concurrently across 12 facilities; Panel B shows the share of survey participants with good cold chain knowledge and attitudes. Data from Asamoah et al. (2021).

Appendix

Material Shortage Risk-Scoring and Mitigation Selection Algorithm

Algorithm A1 operationalizes the proposed framework. It applies the buffering levers that Kouvelis et al. (2023) identify (inventory, time buffers, and supplier flexibility), builds a shortage mechanism into the calculation as Singh et al. (2026) do, and adds the estimation and monitoring safeguards highlighted by Namisango et al. (2017) and Asamoah et al. (2021), while its market-signal inputs reflect the investment concerns reported by Mlyuka et al. (2022). Figure A1 summarizes the steps, and Table A1 and Figure A2 apply the algorithm to illustrative data. The weights are starting assumptions to be calibrated with interview and performance data collected in the proposed study.

Algorithm A1

Material Shortage Risk Scoring (MSRS) for Ramp-Up Planning

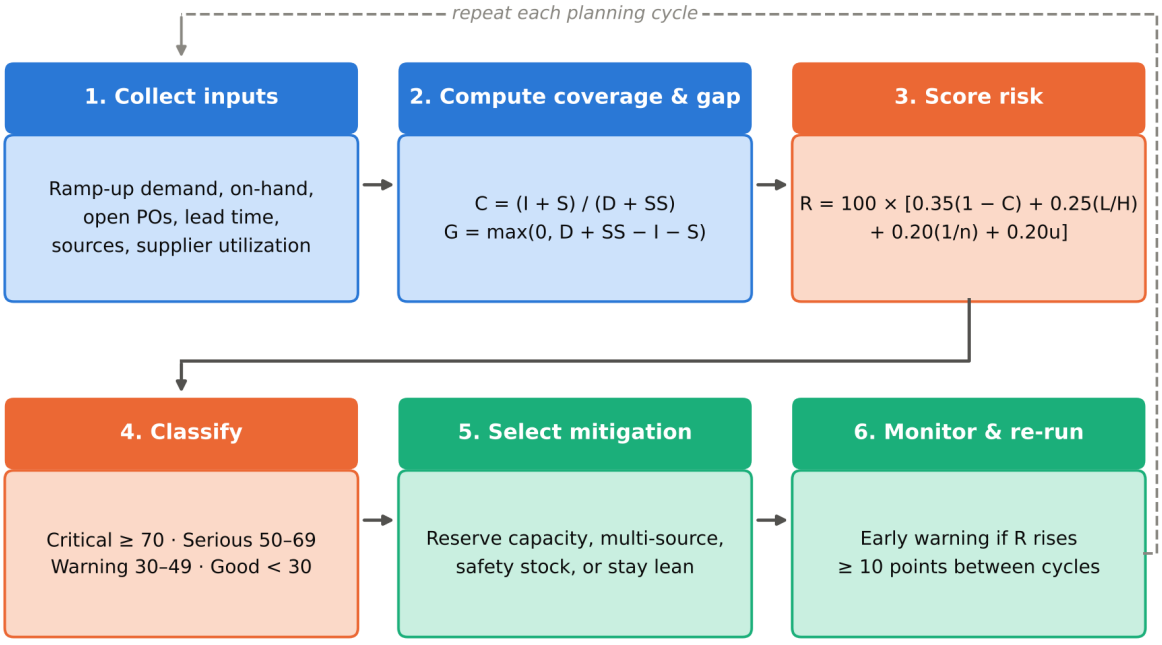
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Input:  for each critical part i in the ramp-up bill of materials:
        D_i  ramp-up demand over horizon H (weeks)
        I_i  on-hand inventory      S_i  confirmed open supply (POs)
        SS_i strategic safety stock L_i  supplier lead time (weeks)
        n_i  qualified sources      u_i  supplier capacity utilization (0-1)
        Weights w = (0.35, 0.25, 0.20, 0.20)
Output: risk score R_i, status, and mitigation action for every part

 1  for each part i do
 2      C_i <- (I_i + S_i) / (D_i + SS_i)                // projected coverage
 3      G_i <- max(0, D_i + SS_i - I_i - S_i)            // shortage gap (units)
 4      V_i <- w1(1 - min(C_i,1)) + w2 min(L_i/H,1)
              + w3(1/n_i) + w4 u_i                      // vulnerability index
 5      R_i <- round(100 * V_i)
 6      if   R_i >= 70 then status <- CRITICAL
 7           action <- reserve capacity, expedite, qualify alternate source
 8      elif R_i >= 50 then status <- SERIOUS
 9           action <- multi-source, raise strategic safety stock
10      elif R_i >= 30 then status <- WARNING
11           action <- weekly supplier capacity review, collaboration
12      else status <- GOOD;  action <- maintain lean / JIT replenishment
13      if R_i - R_i(previous cycle) >= 10 then raise EARLY WARNING
14  end for
15  sort parts by R_i (descending) and escalate CRITICAL parts to the
    cross-functional ramp-up review
16  repeat every planning cycle with refreshed demand and supplier data
```

Note. PO = purchase order; JIT = just-in-time. Developed by the author for the proposed study.

Figure A1

Flow of the Material Shortage Risk-Scoring Algorithm



Note. Blue steps prepare data, orange steps score and classify risk, and green steps act and monitor. C = coverage; G = gap; L/H = lead time relative to the planning horizon; n = qualified sources; u = supplier utilization.

Table A1

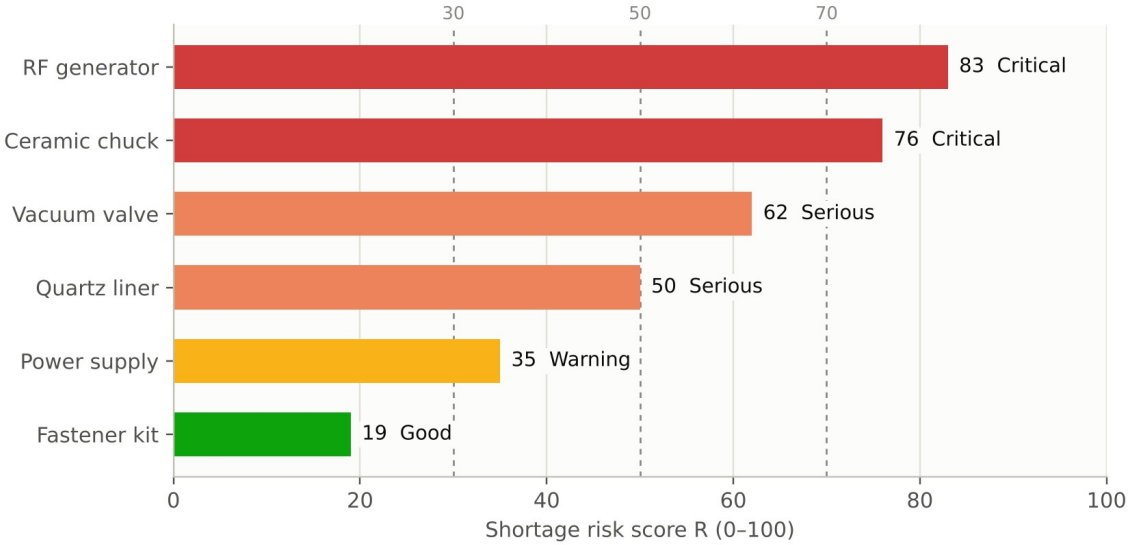
Illustrative Application of Algorithm A1 to Six Ramp-Up Parts (Hypothetical Data)

Part	D	I	S	SS	L	n	u	C	G	R	Status	Mitigation
RF generator	120	20	40	15	26	1	0.95	0.44	75	83	Critical	Capacity reservation, expedite, qualify alternate source
Ceramic chuck	90	15	45	10	20	1	0.85	0.60	40	76	Critical	Capacity reservation, expedite, qualify alternate source
Vacuum valve	300	80	150	30	14	2	0.80	0.70	100	62	Serious	Multi-source and raise strategic safety stock
Quartz liner	200	60	120	20	10	2	0.70	0.82	40	50	Serious	Multi-source and raise strategic safety stock
Power supply	150	70	90	15	8	3	0.60	0.97	5	35	Warning	Weekly supplier capacity review and collaboration
Fastener kit	800	600	400	50	3	4	0.40	1.18	0	19	Good	Maintain lean / just-in-time replenishment

Note. All values are hypothetical and were created only to demonstrate the calculation; they are not data from any firm or study. H = 13 weeks. D = ramp-up demand; I = on-hand; S = confirmed open supply; SS = safety stock; L = lead time in weeks; n = qualified sources; u = utilization; C = coverage; G = gap in units; R = risk score.

Figure A2

Illustrative Shortage Risk Scores by Part (Hypothetical Data)



Note. Dashed lines mark the status thresholds at 30, 50, and 70. Bars are colored by status (red = critical, coral = serious, amber = warning, green = good), and each bar is also labeled with its status so meaning does not depend on color alone. Values are computed from Table A1.

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